20. What is kube-proxy?

Professional Answer

kube-proxy manages networking rules on nodes to enable service-to-pod communication using iptables or IPVS.

Unprofessional Answer

kube-proxy handles networking.

What is the role of kube-proxy in Kubernetes networking?
kube-proxy manages networking rules on worker nodes.
It enables communication between services and pods using iptables or IPVS.
kube-proxy provides service discovery and load balancing functionality.
It ensures traffic sent to a service IP is forwarded to the correct pod.
Without kube-proxy, Kubernetes services would not function correctly.

. kube-proxy (Networking Internals)

What it does internally

- Programs networking rules
- Uses:
    - iptables OR
    - IPVS

Service Traffic Flow

Client → Service IP → kube-proxy rules → Pod IP

Modes

- userspace (old)
- iptables (default)
- IPVS (high scale)

Without kube-proxy

Services won't work

Excellent 👍 — **kube-proxy** is a **very common but deeply misunderstood interview topic**, especially for **networking + troubleshooting rounds**.

I'll cover:

✅ Interview Questions & Answers
✅ Troubleshooting Steps
✅ Real Scenario-Based Questions
✅ Advanced / Senior-Level Discussion
✅ Professional vs Weak Answers (where useful)

Kubernetes kube-proxy – Interview Q&A + Troubleshooting + Scenarios

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PART 1 Basic & Intermediate Interview Questions

Q1. What is kube-proxy?

✅ **Professional Answer**

kube-proxy is a network component that runs on every worker node and manages Service networking.
It maintains network rules that allow Pods to communicate with Services.
It performs load balancing across backend Pods.
kube-proxy translates Service IPs into Pod IPs using iptables or IPVS.
It ensures stable networking abstraction in Kubernetes.

❌ **Weak Answer**

kube-proxy is used for networking between pods.

Q2. Why is kube-proxy needed?

✅ **Professional Answer**

Pods are ephemeral and their IPs change frequently.
kube-proxy provides a stable Service IP abstraction.
It dynamically updates routing rules when Pods are added or removed.
This allows applications to communicate reliably without knowing Pod IPs.
It enables service discovery and load balancing.

Q3. Where does kube-proxy run?

✅ **Professional Answer**

kube-proxy runs as a DaemonSet on every worker node.
Each instance manages networking rules locally on that node.
It communicates with the API Server to watch Service and Endpoint changes.
It updates local routing rules accordingly.

Q4. What are kube-proxy modes?

✅ **Professional Answer**

kube-proxy supports three main modes:

1. User-space mode (deprecated)
2. iptables mode (default)
3. IPVS mode (high performance)

IPVS provides better scalability and load balancing algorithms.

Q5. How does kube-proxy perform load balancing?

✅ **Professional Answer**

kube-proxy distributes traffic across backend Pods.
In iptables mode, it uses NAT rules and round-robin balancing.
In IPVS mode, it uses kernel-level virtual server load balancing.
Traffic is redirected from Service IP to Pod IP.
The load balancing happens at node level.

Q6. What is the difference between iptables and IPVS mode?

Feature	iptables	IPVS
Performance	Medium	High
Scalability	Limited	Excellent
Load Balancing Algorithms	Basic	Multiple algorithms

Q7. Does kube-proxy handle Pod-to-Pod communication?

✅ **Professional Answer**

No. Pod-to-Pod networking is handled by the CNI plugin.
kube-proxy manages Service-level networking.
It handles ClusterIP, NodePort, and LoadBalancer Services.
CNI ensures Pod connectivity.
kube-proxy ensures Service abstraction.

PART 2 Advanced Interview Questions

Q8. How does kube-proxy know when Pods change?

✅ **Answer**

kube-proxy watches the Kubernetes API Server.
It listens for Service and Endpoint updates.
When changes occur, kube-proxy updates routing rules.
This happens dynamically and automatically.
This event-driven model ensures real-time updates.

Q9. What happens if kube-proxy fails?

✅ **Answer**

Existing Pod networking continues to work.
Service networking may break.
New routing rules are not created.
Traffic to Service IPs may fail.
Pod-to-Pod communication still works.

Q10. Can Kubernetes run without kube-proxy?

✅ **Answer**

Yes, if replaced by alternative networking solutions like eBPF-based networking (Cilium).
Traditional Kubernetes Service routing requires kube-proxy.
Modern clusters sometimes replace kube-proxy for performance.
However, kube-proxy is still widely used.
Removing kube-proxy requires compatible networking tools.

PART 3 Troubleshooting kube-proxy (Step-by-Step)

Problem 1: Service Not Reachable

Step 1: Check kube-proxy Pod

kubectl get pods -n kube-system | grep kube-proxy

Step 2: Check Logs

kubectl logs <kube-proxy-pod> -n kube-system

Step 3: Verify iptables/IPVS Rules

iptables -L -n

ipvsadm -L

Step 4: Validate Endpoints

kubectl get endpoints <service>

Step 5: Restart kube-proxy

kubectl rollout restart ds kube-proxy -n kube-system

Problem 2: Service Works From Some Nodes Only

Possible Causes

- kube-proxy crash on specific node
- iptables corruption
- Node network issue

Problem 3: Load Balancing Not Even

Troubleshooting

- Check Service session affinity
- Validate IPVS algorithm
- Check backend Pod readiness

PART 4 Real Scenario-Based Interview Questions 🔥

Scenario 1

Question:

Pods communicate successfully using Pod IP, but Service IP fails. Why?

Expected Answer:

- kube-proxy failure
- iptables corruption
- Endpoints missing
- kube-proxy not synced with API server

Scenario 2

Question:

NodePort Service works externally, but not internally from Pods.

Expected Thinking:

- Hairpin NAT issue
- kube-proxy configuration problem
- Network plugin limitation

Scenario 3

Question:

Traffic goes only to one Pod even though multiple replicas exist.

Expected Answer:

- Session affinity enabled
- Readiness probes failing
- IPVS algorithm configuration

Scenario 4

Question:

After upgrading cluster, Services stopped working cluster-wide.

Expected Answer:

- kube-proxy version mismatch
- iptables vs IPVS mode mismatch
- Kernel module compatibility issue
- CNI plugin incompatibility

Scenario 5

Question:

Service endpoints exist, kube-proxy is running, but Service still unreachable.

Expected Thinking:

- iptables chain corruption
- Node firewall blocking rules
- Kernel networking module issue

PART 5 Senior-Level Deep-Dive Questions

Q: Why do large clusters prefer IPVS over iptables?

Answer:

iptables scales poorly with thousands of rules.
IPVS provides kernel-level load balancing.
It supports multiple scheduling algorithms.
It reduces latency and CPU usage.
IPVS improves cluster networking performance.

Q: How do you monitor kube-proxy health in production?

Answer:

- Monitor kube-proxy logs
- Track Service latency

- Observe packet drops
- Monitor iptables rule count
- Use Prometheus metrics

PART 6 Golden Interview Summary ⏳

👉 kube-proxy manages **Service networking**
👉 CNI manages **Pod networking**
👉 kube-proxy failure affects **Service routing only**
👉 IPVS mode is preferred for **high scale clusters**

⭐ **Killer One-Line Answer**

kube-proxy ensures that Kubernetes Services can route traffic reliably to backend Pods by maintaining dynamic network routing rules.